

Transfer Learning via Fine-tuning for Image Classification

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I. INTRODUCTION

This report presents the experiments conducted on transfer learning and fine-tuning for image classification using different convolutional neural networks. The goal of this work was to investigate how pre-trained models can be adapted to new tasks and to analyze the impact of different fine-tuning strategies, backbone architectures, and classifier head designs.

Overall, the experiments confirm the effectiveness of transfer learning, since pre-trained models converge faster and achieve higher accuracy compared to training from scratch. Full fine-tuning typically provides the best performance in our case, as it allows the network to adapt high-level features to the target dataset. Moreover it's showed how increasing the complexity of the model or of the head of the model doesn't bring to significant improvements generally, but can make a difference for some particular classes.

II. METHOD

All experiments were conducted on the CIFAR-10 dataset. Images were converted to tensors and normalized using the mean and standard deviation of the ImageNet dataset in order to match the statistics used during the pre-training of the backbone networks.

A. Transfer learning vs training from scratch

We evaluated the performance of the ResNet50 model on CIFAR-10 with and without ImageNet's pre-trained weights. Different fine-tuning strategies were explored:

- **Training from scratch:** the network weights were randomly initialized and trained entirely on CIFAR-10.
- **Feature extraction:** the backbone parameters were frozen and only the final classifier layer was trained.
- **Selective fine-tuning:** the last block of the backbone network was unfrozen while earlier layers remained fixed.
- **Full fine-tuning:** all network parameters were updated during training.

These strategies allow analyzing the trade-off between preserving generic features learned from large datasets and adapting the model to the target task.

B. Reduced dataset and backbone comparison

We experimented using a reduced version of the CIFAR-10 dataset containing only four classes: *cat*, *dog*, *bird*, and *airplane*. This subset was selected to analyze the behavior of transfer learning methods in a more controlled setting. Using fewer classes also reduces computational cost and allows faster experimentation with different architectures,

so we tested both a lightweight backbone (ResNet18) and a heavyweight backbone (ResNet101), both with the same fine-tuning method. The chosen classes include both visually similar categories (cat and dog) and more distinct ones (bird and airplane), providing a balanced scenario to evaluate the model's ability to distinguish between both subtle and clear visual differences. Moreover, we wanted to observe if the difference between the scenarios in which the two couples of classes are likely to be showed would be relevant in the models' performances.

C. Classifier Head Design

In the final experiment, two different classifier head architectures were compared on the reduced dataset while keeping the backbone and the fine-tuning strategy fixed. A simple linear classifier was used as a baseline, while an alternative head included additional fully connected layers and nonlinear activation functions. This experiment aimed to evaluate whether increasing the complexity of the classification head improves the adaptation of pre-trained features to the target dataset.

III. IMPLEMENTATION

All experiments were implemented in Python using the PyTorch deep learning framework. The code is organized in three Jupyter notebooks corresponding to the three experimental tasks. All the models were imported through the `torchvision.models` library, which offers the possibility to set ImageNet's pre-trained weights.

In this section we briefly explain the technical implementation of the most relevant parts of the experiments.

A. Transfer learning vs training from scratch: Freeze mode

In the first experiment, we built a `train_setup(freeze_mode='')` function that can setup a pre-trained ResNet50 model with one of the three different fine-tuning modalities. The freeze mode is implemented by setting `param.requires_grad = False/True` for the different parameters:

- `freeze_mode='feature_extraction'` implements feature extraction
- `freeze_mode='selective_fine_tuning'` implements selective fine tuning
- Full fine-tuning is implemented by default

The function returns the model setted and loaded on the device, the optimizer, the criterion and the dataset loaders.

B. Reduced dataset and backbone comparison

The dataset is reduced using the `select_classes_dataset` function, then we trained both pre-trained ResNet18 and ResNet101 on the reduced dataset using full fine-tuning.

C. Classifier Head Design

We trained pre-trained ResNet18 using full-fine tuning both with a linear classifier head (the same from the precedent experiments) and with a deeper one, implemented by the `nn.Sequential` function. In particular, the structure for the new classifier head is Linear $\rightarrow$ ReLU $\rightarrow$ Dropout $\rightarrow$ Linear.

IV. DATA

In this section we show the relevant results obtained in each experiment.

A. Transfer learning vs training from scratch

• Training from scratch

- Final test accuracy: 69.10%
- Per-class accuracies
 - * airplane: 72.20%
 - * automobile: 67.70%
 - * bird: 52.10%
 - * cat: 55.50%
 - * deer: 50.30%
 - * dog: 64.60%
 - * frog: 80.30%
 - * horse: 78.60%
 - * ship: 87.30%
 - * truck: 82.40%

• Feature extraction

- Final test accuracy: 47.35%
- Per-class accuracies
 - * airplane: 56.40%
 - * automobile: 52.80%
 - * bird: 37.80%
 - * cat: 38.40%
 - * deer: 42.10%
 - * dog: 37.70%
 - * frog: 51.40%
 - * horse: 44.80%
 - * ship: 55.80%
 - * truck: 56.30%

• Selective fine-tuning

- Final test accuracy: 77.11%
- Per-class accuracies
 - * airplane: 79.60%
 - * automobile: 81.50%
 - * bird: 72.40%
 - * cat: 60.30%
 - * deer: 75.10%
 - * dog: 74.30%
 - * frog: 81.40%
 - * horse: 79.20%

- * ship: 87.20%
- * truck: 80.10%

• Full fine-tuning

- Final test accuracy: 84.46%
- Per-class accuracies
 - * airplane: 89.10%
 - * automobile: 89.50%
 - * bird: 85.50%
 - * cat: 71.30%
 - * deer: 84.30%
 - * dog: 66.50%
 - * frog: 89.20%
 - * horse: 91.30%
 - * ship: 92.40%
 - * truck: 85.50%

B. Reduced dataset and backbone comparison

• Lightweight backbone: ResNet18

- Number of parameters: 11178564
- Training time: 103s
- Final test accuracy: 80.72%
- Per-class accuracies
 - * dog: 76.90%
 - * cat: 70.00%
 - * airplane: 93.70%
 - * bird: 82.30%

• Heavyweight backbone: ResNet101

- Number of parameters: 42508356
- Training time: 363s
- Final test accuracy: 76.92%
- Per-class accuracies
 - * dog: 69.30%
 - * cat: 73.50%
 - * airplane: 92.10%
 - * bird: 72.80%

C. Classifier Head Design

• Linear Head

- Final test accuracy: 81.47%
- Per-class accuracies
 - * dog: 78.00%
 - * cat: 77.30%
 - * airplane: 90.70%
 - * bird: 79.90%

• Deeper Head

- Final test accuracy: 80.67%
- Per-class accuracies
 - * dog: 68.10%
 - * cat: 75.00%
 - * airplane: 93.30%
 - * bird: 86.30%

V. RESULTS AND ANALYSIS

A. Transfer learning vs training from scratch

Among the evaluated approaches, pre-trained models converged faster than training from scratch, confirming the

advantage of transfer learning when limited training data is available. Feature extraction showed the lowest performance (47.35% accuracy), indicating that frozen ImageNet features alone are not sufficient to fully adapt to CIFAR-10. In contrast, full fine-tuning achieved the best generalization (84.46%), since all layers could adapt to the new domain.

The classes that benefited the most from pretraining were airplane, ship, horse and frog, which reached accuracies close to or above 90% with full fine-tuning. These categories likely benefit from features already learned on ImageNet such as edges, shapes and textures. More difficult classes such as cat and dog showed smaller improvements, probably due to the higher visual similarity between them and the domain gap between ImageNet and CIFAR-10 images.

B. Reduced dataset and backbone comparison

Increasing the backbone size did not improve performance in our experiments. The lighter ResNet18 achieved a higher accuracy (80.72%) compared to the larger ResNet101 (76.92%), while also requiring significantly less training time. The improvement was not consistent across classes: ResNet101 slightly improved the cat class, but performed worse on dog and bird.

These results highlight the trade-off between model capacity and computational cost. Larger models contain more parameters, so they require longer training times and more data: they do not necessarily generalize better on relatively small datasets such as the reduced CIFAR-10 subset.

C. Classifier Head Design

Adding a deeper classifier head did not significantly improve overall performance. The simple linear head achieved a slightly higher accuracy (81.47%) compared to the deeper head (80.67%). The deeper head improved some individual classes such as bird and airplane, but degraded performance on dog.

This suggests that when the backbone is already fully fine-tuned, most of the task-specific adaptation occurs in the convolutional layers. As a result, increasing the complexity of the classifier head provides limited benefits, since the backbone already extracts sufficiently discriminative features.

An interesting observation concerns the large performance gap between classes in the last two experiments. In both, the airplane class consistently achieved the highest accuracy. This is likely due to the distinctive visual structure of airplanes, which typically appear with clear shapes and relatively uniform backgrounds (e.g., sky), making them easier to recognize using high-level features learned during pretraining.

In contrast, the bird class shows more variability in pose and scale, while probably maintaining a similar background, which makes it more difficult to classify. In Experiment B, this variability likely affected the larger backbone more strongly, leading to a drop in accuracy. However, in Experiment C the deeper classifier head slightly improved the performance on "flying" classes in general, while the "earth"

classes' accuracies decreased, suggesting that the additional nonlinear layers may help capture features related to more distinctive shapes but may introduce unnecessary complexity for already separable classes.

VI. CONCLUSIONS

In this work we evaluated the impact of transfer learning, backbone architecture, and classifier head design on image classification performance. The experiments showed that using pretrained models significantly improves convergence speed and final accuracy compared to training from scratch, with full fine-tuning providing the best results. Increasing the backbone complexity did not lead to better performance on the reduced dataset, highlighting the trade-off between model capacity and computational cost. Finally, modifying the classifier head had only a limited impact, suggesting that most task-specific adaptation occurs in the backbone features.

VII. TIME LOG

- Andrea Sabbatini
 - Coding: 2h
 - Test: 1h
 - Writing: 3h
- Adham HAMDY
 - Coding: 2h
 - Test: 2h
 - Writing: 1h
- Alexandra
 - Coding: 2h
 - Test: 1h
 - Writing: 1h

REFERENCES

- [1] Kaiming He, Xiangyu Zhang, Shaoqing Ren, Jian Sun, Deep Residual Learning for Image Recognition, 2016 IEEE Conference on Computer Vision and Pattern Recognition